Table 1: Optimization variables (`opti.variable`) with bounds.

Variable	Init. guess	Lower	Upper	Units	Description
Wing					
`S`	0.2	0.01	1	m^2	Reasonable Bounds
`AR`	3	2	15	–	Reasonable Bounds
Tail - goal is for preliminary mass model, not stability. Stability will be checked later with AVL					
`AR_v`	2	1	5	–	Reasonable - will check stability with AVL
`AR_h`	4	3	5	–	Reasonable - will check stability with AVL
`S_v`	0.01	0.001	0.1	m^2	Reasonable - will check stability with AVL
`S_h`	0.01	0.001	0.1	m^2	Hitting the upper bound
`x_tail`	1	0.7	1.2	m	Hitting the upper bound
Sensor and mass					
`payload_mass_frac`	10	0	5	–	Payload mass / Plane empty mass - reasonable bounds
`mass_sensor`	1.5	0	3	kg	??
`mass_container`	1	0.5`mass_sensor`	none	kg	the box weighs nontrivially, though this is a guess
`sensor_height`	0.12	0.127 (3 in)	0.254 (10 in)	m	a guess - reasonable fineness ratio - check hoerner
`sensor_width`	0.1	0.0762 (3 in)	0.1524 (6 in)	m	a guess
`sensor_length`	0.3048 (12 in)	0.254 (10 in)	0.3556 (14 in)	m	more guesswork!
Component locations - These can mostly go whereever they want, important for the Aerosandbox native aero and stability checks, not super important					
`x_sensor`	0	−0.1	0.2	m	near the CG
`x_motor`	−0.3	−0.4	−0.2	m	can't go farther than a certain point - set it so it would stay in front. Kept there for future weight balancing CG purposes.
`x_electronics`	−0.1	−0.5	$0.8\,c_r$	m	
`x_battery`	−0.1	−0.5	$0.8\,c_r$	m	
`x_battery_avionics`	−0.2	−0.5	c_r	m	
Operating point (one per mission, M2 and M3)					
`velocity[M2]`, `velocity[M3]`	20	0	–	m/s	
`alpha[M2]`, `alpha[M3]`	0	−10	13	deg	Not that important for initial optimization sizing.
`M3_duration`	300	0	300	s	Max 5 mins - rules.

Table 2: Set (fixed) variables and constants.

Variable	Value	Units	Description
Configuration			
`given_airfoil`	False	–	For the optimizer
`mission_no`	0	–	Mission dependent optimization
`missions`	(M2, M3)	–	Optimizer's internal bookkeeping of the missions.
Wing and tail design			
`taper`	1	–	Unnecessary to taper - no significant efficiencies gained.
`airfoil`	sd7032	–	Best cd0 from CL 0.3-0.6.

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Table 2 (continued)

Variable	Value	Units	Description
`airfoil_tail`[1]	naca0012	–	Classic airfoil, easy to choose, thick enough for structural stability.
`dihedral`	0	deg	0 for ease of building, good enough tail should be stable enough to not increase complexity.
`washout`	0	deg	Unnecessary because rectangle wing won't stall at the tip.
`decalage`	-5	deg	suboptimal
`flap hinge_point`	0.6	–	Need to run AVL to ensure authority.
`aileron hinge_point`	0.7	–	Need to run AVL to ensure authority.
`elevator hinge_point`	0.70	–	Need to run AVL to ensure authority.
`rudder hinge_point`	0.75	–	Need to run AVL to ensure authority.
Aerodynamics and performance			
`rho`[2]	1.10	kg/m^3	In notes
`nu`[3]	1.506×10^{-5}	m^2/s	In notes
`g`	9.81	m/s^2	Constant
`N_max`[4]	10	–	needs justification
`CL_max`[5]	1.4	–	why are they different
`stall_CL`[6]	1.5	–	justification
`stall_speed`[7]	15	m/s	justification
`interference_factor`[8]	1.5	–	Hoerner - check
`CDp`	0.02	–	SD7032 CD0 - should check
`e`[9]	0.9	–	check avl
`CD_lg`[10]	0.25	–	find equation
`C_f`[11]	0.004	–	find equation
`S_g`[12]	91.44 (300 ft)	m	why choose this distance
`area_chord_tail` [13]	0.0822	–	Do the math (asked chat)
Mission			
`lap_dist`	1000	m	relatively generous estimate assuming cruise only
`M2_max`	0.065	–	pmo
`M3_max`	35	–	pmo
Propulsion			
`battery_power`	100×3600	J	rules, max battery power
`voltage`	25	V	8s - change to 29.6/figure out battery sag
`max_amperage`	100	A	rules, max amperage
`safety_factor`[14]	1.5	–	Arbitrary
Inline literals in the code (suggested names)			
`eta_prop`[15]	0.7	–	arbitrary/needs testing
`empty_mass_factor`	1.3	–	justify
`deflection_frac`	0.08	–	deflection percentage, justify
Sensor			
`ratio_container_sensor`	0.5	–	sigh - random numbers
Component masses			
`mass_electronics`	0.1915	kg	check the numbers
`mass_battery_avionics`	0.07	kg	check the numbers
`mass_propeller`	0.56	kg	prop and motor
`mass_tail_servo`	0.07	kg	check the numbers
`mass_wing_servo`	0.14	kg	check the numbers
`mass_landing_gear`	0.2	kg	check the numbers
`mass_misc`	2.0	kg	check the numbers
`mass_spar_misc`	0.009	kg	check the numbers
Spar geometry			

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Table 2 (continued)

Variable	Value	Units	Description
`spar_cap_thickness`	0.002	m	?? LOL
`spar_wood_width`	0.009525 (3/8 in)	m	Existing material
Carbon fiber			
`carbon_fiber_density`	1750	kg/m^3	
`carbon_fiber_yield_strength`	600×10^6	Pa	
`carbon_fiber_youngs_modulus`	230×10^9	Pa	
`carbon_fiber_shear_modulus`	50×10^9	Pa	
`carbon_fiber_layup_epoxy_factor`	2.2	–	
`carbon_fiber_areal_density`	0.2	kg/m^2	
Fiberglass			
`fiberglass_density`	2600	kg/m^3	
`fiberglass_areal_density`	0.017	kg/m^2	
`fiberglass_youngs_modulus`	70×10^9	Pa	
`fiberglass_shear_modulus`	30×10^9	Pa	
`fiberglass_layup_epoxy_factor`	2.2	–	
`fiberglass_thickness`	0.0001	m	
Kevlar			
`kevlar_density`	1400	kg/m^3	
`kevlar_layup_epoxy_factor`	1	–	
`kevlar_youngs_modulus`	100×10^9	Pa	
`kevlar_yield_strength`	3×10^9	Pa	
`kevlar_shear_modulus`	717×10^6	Pa	
Basswood			
`basswood_density`	320	kg/m^3	
`basswood_youngs_modulus`	10×10^9	Pa	
`basswood_shear_modulus`	3.8×10^9	Pa	
`basswood_layup_epoxy_factor`	1.2	–	
`basswoord_shear_strength`	6.8×10^6	Pa	
Plywood, balsa, foam, film			
`plywood_density`	550	kg/m^3	
`plywood_thickness`	0.0025	m	
`balsa_density`	160	kg/m^3	
`balsa_youngs_modulus`	4×10^9	Pa	
`balsa_shear_modulus`	1.5×10^9	Pa	
`balsa_layup_epoxy_factor`	1.2	–	
`balsa_shear_strength`	2.1×10^6	Pa	
`foam_density`	25	kg/m^3	
`monokote_density`	0.06	kg/m^2	

Table 3: Constraints (`opti.subject_to`). "AF" = the `given_airfoil=True` (AeroBuildup/AVL) path; "no AF" = the reduced-order path.

Constraint	Applies to	Description
Geometry		
$b < 5.95$ ft	all	slight manufacturing safety margin, but rules.
$c_v = c_h$	all	manufacturing ease/not that critical
$x_{\text{tail}} > c_r$	all	makes sure the plane is a plane
$0.3 < V_h < 0.6$	all	Tail coefficients
$0.02 < V_v < 0.05$	all	Tail coefficients
Mass and sensor		

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Table 3 (continued)

Constraint	Applies to	Description
$m_{\text{empty}} > 1.3\, m_{\text{total}}$	all	ensuring the empty mass > 1.3 times the mass total, though the 1.3 is arbitrary
$m_{\text{mission}} < 55$ lb	each mission	Rules
$\frac{m_{\text{sensor}} + m_{\text{container}}}{V_{\text{sensor}}} < 3000$ kg/m^3	all	can't be denser than steel
Aerodynamics		
$C_L < 0.5$	each mission	kind of arbitrary, not hitting against, could remove
$L = m_{\text{mission}}\, g$	each mission, AF only	cruise equilibrium
$tw_{\text{to}} < 0.9$	M2	thrust to weight ratio at take off - should check
$\frac{1}{2}\, V_{\text{stall}}^2\, \rho\, S\, C_{L,\text{stall}} > m_{M2}\, g$	M2	stall speed is safe
Stability (AF only)		
SM $= 0.15$	each mission	static margin = 0.15 - classic stability
$0.04 < C_{n\beta,\text{corr}} < 0.1$	each mission	more stability
$-1.8 < C_{m\alpha,\text{corr}} < -0.8$	each mission	more stability
Propulsion		
$I_{\max}\, V > P_{\text{mission}}$	each mission	checking power consumption during cruise
$P_{\text{mission}}\, t_{\text{mission}} < E_{\text{avail}}$	each mission	ensuring each mission has enough power
$t_{M2} < 300$ s	M2	making sure M2 occurs in under 5 minutes
Structures		
$\delta_{M2} < \delta_{\max} = 0.08\,(b/2)$	M2	8% wing tip deflection - needs justifying.

Notes

[1] `airfoil_tail`: **[TODO: add note]**
[2] `rho`: Air density used for all missions. Citation: `https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/earth-atmosphere-equ ?utm_source=chatgpt.com` Tuscan height above sea level is 800m, T = 9.8K, P = 92.0kPa –> $\rho = 1.1 kg/m^3$
[3] `nu`: Calculation for kinematic viscosity: Citation: `https://www.engineeringtoolbox.com/air-absolute-kinematic-viscosity-d_ 601.html?vA=22&units=C#` Used 20C
[4] `N_max`: **[TODO: add note]**
[5] `CL_max`: **[TODO: add note]**
[6] `stall_CL`: **[TODO: add note]**
[7] `stall_speed`: **[TODO: add note]**
[8] `interference_factor`: **[TODO: add note]**
[9] `e`: **[TODO: add note]**
[10] `CD_LG`: **[TODO: add note]**
[11] `CF`: **[TODO: add note]**
[12] `S_g`: **[TODO: add note]**
[13] `area_chord_tail`: **[TODO: add note]**
[14] `safety factor`: **[TODO: add note]**
[15] `eta_prop`: **[TODO: add note]**